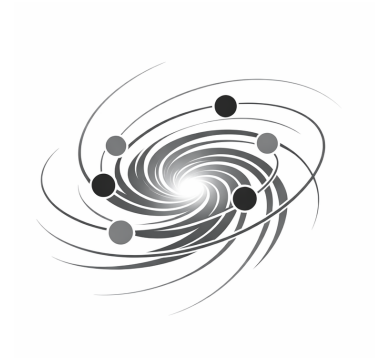


Part of the Monistic Continuum Model (MCM)



Planck–Vortex Theory (PVT)

Emergent Geometry of a Single Medium

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1 Definition of the PVT

The Planck–Vortex Theory (PVT) describes the conditions under which the internal stress, rotation, and curvature geometry of the medium forms stable bound configurations. After the emergence of space, this bound geometry appears as an epistemic material structure. The PVT is a purely geometric binding theory within a spaceless medium and uses exclusively ontological quantities of the Monistic Continuum Model (MCM).

A Planck–Vortex is the only structure that satisfies all stability conditions of the medium. It arises through closed rotation, coherent curvature, and stable stress organization. The PVT defines vortices as the fundamental bound process geometry of the medium.

Phase Structure of the Medium and Phase-1 Islands

The PVT operates exclusively within the active Phase 3 of the medium. The Monistic Continuum Model (MCM) distinguishes three fundamental phases of the medium state, which determine the existence conditions of bound geometry:

- **Phase 1:** State of minimal reaction speed. The medium is fully passive and forms the ontological basis for all higher phases. Phase 1 carries passive Phase 3 bodies, referred to as *Phase-1 islands*.
- **Phase 2:** Transition zone between passive and active medium states. In this zone, vortex spaces emerge or collapse. It is dynamic but not stable.
- **Phase 3:** State of maximal reaction speed. All active vortex spaces and bound structures exist here. The PVT describes exclusively the internal geometry of this phase.

Phase-1 Islands Phase-1 islands are passive Phase 3 bodies whose internal vortex coupling has been completely destroyed. They possess material visibility, mass, and form, but no internal dynamics. Ontologically, they rest directly on Phase 1 and are supported by its stability. Examples include the Moon and other large passive bodies. Phase-1 islands are therefore the only Phase 3 structures without internal vortex activity.

The PVT treats Phase-1 islands as boundary cases of bound geometry: they are geometrically closed but dynamically neutral. Their existence marks the transition between active vortex geometry (Phase 3) and the passive medium state (Phase 1).

Axioms and Fundamental Assumptions

The PVT is based on the ontological assumptions of the Monistic Continuum Model (MCM). The medium is the only fundamental entity. All structures are internal geometric relations of the medium.

The state of the medium is described by the pre-spatial mapping

$$C(\alpha, \Omega) = (\rho, v_A, \sigma_{AB}, \tau)$$

where α is a medium index and Ω is a global update index.

The PVT introduces no physical fundamental entities. Space, time, fields, particles, and forces arise solely through bound geometry and possess epistemic status. Binding is a stability phenomenon of the medium and occurs exactly when internal updating is locally neutral:

$$DC(\alpha, \Omega) \approx 0.$$

Axioms of the PVT

Axiom PVT-1: Stress Dominance Bound geometry arises exclusively through internal stress organization:

$$\sigma_{AB} \rightarrow \gamma_{AB} \rightarrow \mathcal{R}_{ABCD}.$$

Axiom PVT-2: Rotational Binding Stable vortices require closed rotation:

$$L_{AB} \neq 0, \quad \nabla^A L_{AB} = 0.$$

Axiom PVT-3: Curvature Stability Bound geometry requires coherent internal curvature:

$$\mathcal{R}_{ABCD} \text{ stable over a binding region.}$$

Axiom PVT-4: Update Neutrality Stability arises only when

$$DC(\alpha, \Omega) \approx 0.$$

Axiom PVT-5: Pre-Spatial Locality Binding occurs exclusively in the medium-index space:

$$\alpha \in \mathcal{A}.$$

Axiom PVT–6: Vortex as the Only Stable Structure The only structure that satisfies all conditions is the Planck–Vortex:

$$W(\alpha, \Omega) = (\sigma_{AB}, L_{AB}, \mathcal{R}_{ABCD})_{\text{bound}}.$$

Operator Consistency

The PVT uses exclusively geometric operators of the medium. All operators act on medium indices and do not assume physical spacetime.

Stress Operator

$$\mathcal{S}[\sigma_{AB}].$$

Rotation Operator

$$\mathcal{L}[v_A].$$

Curvature Operator

$$\mathcal{R}[\gamma_{AB}].$$

Update Operator

$$D[C].$$

All equations must be geometrically closed. Physical operators of QM, QFT, or GR must not be introduced fundamentally. Correspondences are purely interpretative mappings.

Topological Consistency

The PVT requires an internal topology of the medium that enables binding. Rotation must form a closed loop, stress must generate a coherent domain, and curvature must remain stable across the binding region.

Topological Binding Condition

$$\oint L_{AB} d\alpha^A = n \Phi_0, \quad n \in \mathbb{Z}.$$

The topology must not assume external spaces. All structures are purely internal relations of the medium.

Dynamic Consistency

The dynamics of the PVT are entirely internal. The internal evolution of the medium is described by

$$DC(\alpha, \Omega)$$

Stability

$$DC(\alpha, \Omega) = 0$$

defines stable bound geometry.

Instability

$$DC(\alpha, \Omega) \neq 0$$

leads to open or unstable bound geometry.

Binding Dynamics

$$D\sigma_{AB} = F(Vv, \sigma)_{AB}, \quad Vv(\sigma_{AB}) = v_C \nabla_C \sigma_{AB}.$$

Binding arises through self-organization of the stress field.

2 Mathematical Foundations

The mathematical foundation of the PVT is based on the pre-spatial geometry of the medium. All structures are defined through internal relations between stress, rotation, curvature, and update dynamics. The mathematical framework is entirely spaceless and uses only medium indices. It possesses no physical fundamental status.

Mathematical Carrier Space

The carrier space of the PVT is the medium-index space

$$\alpha \in \mathcal{A},$$

which has no metric, topological, or spatial structure. All geometric quantities are functions over this index space.

The pre-spatial state mapping is given by

$$C(\alpha, \Omega) = (\rho, v_A, \sigma_{AB}, \tau).$$

The carrier space is a purely geometric category without physical interpretation. Space emerges only through bound geometry.

Fundamental Operators

The PVT uses exclusively geometric operators of the medium.

Stress Operator

$$\mathcal{S}[\sigma_{AB}] = \gamma_{AB}.$$

Rotation Operator

$$\mathcal{L}[v_A] = \frac{1}{2}(\nabla_A v_B - \nabla_B v_A).$$

Curvature Operator

$$\mathcal{R}[\gamma_{AB}] = R_{ABCD}.$$

Update Operator

$$D[C] = DC(\alpha, \Omega).$$

All operators act exclusively on medium indices and possess no physical status.

Spectral Theory

The spectral theory of the PVT describes the eigenstructures of the internal geometry of the medium. For every bound configuration, eigenvalues and eigenmodes of the stress, rotation, and curvature geometry exist.

Spectral Equation

$$\mathcal{O}[\Psi_n] = \lambda_n \Psi_n,$$

where $\mathcal{O}$ is a geometric operator of the medium.

Spectral Stability Stable bound geometry requires a discrete spectrum:

$$\lambda_n \in \mathbb{R}_{\text{discrete}}.$$

Unstable structures possess a continuous spectrum.

Phase Dependence of Mathematical Operators

The mathematical operators of the PVT act exclusively within the active Phase 3 of the medium. Phase 1 and Phase 2 possess no stable operator space, as no closed vortex coupling exists there. The operators $\mathcal{S}$, $\mathcal{L}$, $\mathcal{R}$, and D are therefore valid only in regions where the internal updating of the medium is locally neutral:

$$DC(\alpha, \Omega) \approx 0.$$

Phase Boundary Conditions

- In **Phase 1**, $DC(\alpha, \Omega) = 0$ is trivial, since no internal dynamics exist. Operators are ineffective here.
- In **Phase 2**, $DC(\alpha, \Omega) \neq 0$, causing operators to become unstable and preventing closed geometry.
- In **Phase 3**, $DC(\alpha, \Omega) = 0$ acts as a stability condition. Only here do bound vortex spaces and stable eigenstructures arise.

Phase-1 Islands as a Special Case Phase-1 islands are geometrically bound but dynamically neutral configurations within Phase 3 that rest directly on Phase 1. Mathematically, they possess constant operator values:

$$\mathcal{S}[\sigma_{AB}] = \text{const}, \quad \mathcal{L}[v_A] = 0, \quad \mathcal{R}[\gamma_{AB}] = \text{stable}.$$

Thus, Phase-1 islands are the only structures for which all operators are stationary. They form the boundary condition between active vortex space and the passive medium state.

Projection onto Continuous Fields

After the emergence of space, bound geometry is interpreted as a field-like structure. The projection is given by the mapping

$$x = X(\alpha, \Omega),$$

where x is defined only after stabilization.

The emergent metric is

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

All field structures are projections of bound geometry and possess epistemic status.

Stability Norms

Stability of bound geometry is defined through norms of the internal operators.

Stress Norm

$$\|\sigma\|^2 = \sigma_{AB}\sigma^{AB}.$$

Rotation Norm

$$\|L\|^2 = L_{AB}L^{AB}.$$

Curvature Norm

$$\|\mathcal{R}\|^2 = R_{ABCD}R^{ABCD}.$$

Stability Condition

$$DC(\alpha, \Omega) = 0.$$

Emergent Physical Objects

Physical objects arise only after the emergence of space. They possess epistemic status and are projections of bound geometry.

Matter Structure

$$M(x, \Omega) = (\sigma_{AB}, L_{AB}, \mathcal{R}_{ABCD})_{\text{bound}}.$$

Matter is an emergent projection of bound geometry.

Quark–Vortex Geometry

Quark–vortices are bound configurations with minimal rotational charge. They possess closed rotation and a locally coherent curvature domain.

Definition

$$W_q(\alpha, \Omega) = (\sigma_{AB}, L_{AB}, \mathcal{R}_{ABCD})_{\text{minimal}}.$$

Quark–vortices are elementary building blocks of higher-order vortex structures.

Higher-Order Vortices

Higher-order vortices arise through the combination of multiple quark–vortex geometries.

Binding Condition

$$\oint L_{AB} d\alpha^A = n \Phi_0.$$

Structure

$$W_n = \sum_{k=1}^n W_q^{(k)}.$$

Vortex Clusters

Vortex clusters are coherent groups of bound vortices. They form the basis of complex bound geometry.

Cluster Definition

$$C_W = \{W_1, W_2, \dots, W_m\}.$$

Clusters possess shared curvature domains.

Stable and Unstable Vortices

Stable

$$DC(\alpha, \Omega) = 0.$$

Unstable

$$DC(\alpha, \Omega) \neq 0.$$

Stable vortices form material appearances. Unstable vortices decay into open or transitional geometry.

3 Geometric Operators

Geometric operators define the internal structure of the medium and the development of bound and unbound geometry. They act exclusively on medium indices and do not assume physical spacetime. All operators are purely geometric constructs of the medium and possess no physical fundamental status.

Topological Operators

Topological operators describe closed rotational structures and binding conditions within the medium.

Rotation Integral

$$\mathcal{T}[L] = \oint L_{AB} d\alpha^A.$$

Topological Charge

$$\mathcal{Q} = \frac{1}{\Phi_0} \oint L_{AB} d\alpha^A, \quad \mathcal{Q} \in \mathbb{Z}.$$

Topological operators define the closed structure of a vortex and possess purely geometric status.

Dynamic Operators

Dynamic operators describe the internal updating of the medium.

Update Operator

$$D[C] = DC(\alpha, \Omega).$$

Transport Operator

$$Vv(\sigma_{AB}) = v_C \nabla_C \sigma_{AB}.$$

Stress Dynamics

$$D\sigma_{AB} = F(Vv, \sigma)_{AB}.$$

Dynamic operators define the evolution of unbound and bound geometry.

Phase Dependence of Geometric Operators

The geometric operators of the PVT act exclusively within the active Phase 3 of the medium. In Phase 1 and Phase 2 no stable vortex spaces exist, and therefore no closed operator action is defined. The operators $\mathcal{S}$, $\mathcal{L}$, $\mathcal{R}$, and D are valid only in regions where the internal updating of the medium remains locally neutral:

$$DC(\alpha, \Omega) \approx 0.$$

Operator Action Across Phases

- **Phase 1:** The medium is fully passive. All operators are stationary because no internal dynamics exist.
- **Phase 2:** Transition zone between passive and active medium states. Operators are unstable here and generate open geometries.
- **Phase 3:** Active vortex space. All operators act coherently and generate bound geometry.

Phase-1 Islands Phase-1 islands are special cases within Phase 3 where operators take stationary values:

$$\mathcal{S}[\sigma_{AB}] = \text{const}, \quad \mathcal{L}[v_A] = 0, \quad \mathcal{R}[\gamma_{AB}] = \text{stable}.$$

They possess no internal dynamics and rest directly on Phase 1. They form the geometric boundary condition between active vortex space and the passive medium state.

Observables

Observables are geometric quantities that appear as measurable structures after the emergence of space. They possess epistemic status and are not physical entities.

Stress Observable

$$\mathcal{O}_\sigma = \sigma_{AB}\sigma^{AB}.$$

Rotation Observable

$$\mathcal{O}_L = L_{AB}L^{AB}.$$

Curvature Observable

$$\mathcal{O}_R = R_{ABCD}R^{ABCD}.$$

From Operators to Geodesics

Geodesics arise only after the emergence of space. They are projections of bound geometry and possess no ontological status within the medium.

Emergent Geodesic

$$\frac{d^2 x^i}{d\lambda^2} + \Gamma_{jk}^i(x, \Omega) \frac{dx^j}{d\lambda} \frac{dx^k}{d\lambda} = 0.$$

Origin of Geodesics

$$\Gamma_{jk}^i(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^j} \frac{\partial X^B}{\partial x^k}.$$

Geodesics are emergent mappings of bound geometry.

Interaction Operators

Interactions arise through the superposition of bound stress, rotation, and curvature geometries. They are internal relations, not forces.

Interaction Operator

$$\mathcal{I}[W_1, W_2] = \sigma_{AB}^{(1)} \sigma^{(2)AB} + L_{AB}^{(1)} L^{(2)AB} + R_{ABCD}^{(1)} R^{(2)ABCD}.$$

Spectral Operators

Spectral operators define the eigenmodes of bound geometry.

Spectral Operator

$$\mathcal{O}[\Psi_n] = \lambda_n \Psi_n.$$

Discrete Spectrum

$$\lambda_n \in \mathbb{R}_{\text{discrete}}$$

for stable vortices.

Continuous Spectrum

$$\lambda \in \mathbb{R}_{\text{continuous}}$$

for unstable structures.

Fundamental Physical Parameters

Physical parameters arise only after the emergence of space and possess epistemic status. They are interpretations of bound geometry.

Emergent Energy

$$E = \mathcal{O}_\sigma + \mathcal{O}_L + \mathcal{O}_R.$$

Emergent Momentum Structure

$$p_i = g_{ij} \frac{dx^j}{d\lambda}.$$

Emergent Charge

$$q = \mathcal{Q}.$$

All parameters are projections of geometric operators.

Fundamental Dynamic Equations

The fundamental equations of the PVT are purely geometric.

Stress Evolution

$$D\sigma_{AB} = F(Vv, \sigma)_{AB}.$$

Rotation Evolution

$$DL_{AB} = G(\sigma, L)_{AB}.$$

Curvature Evolution

$$D\mathcal{R}_{ABCD} = H(\gamma, \mathcal{R})_{ABCD}.$$

These equations define the internal dynamics of the medium.

Vortex Networks

Vortex networks are coherent groups of bound vortices.

Network Definition

$$\mathcal{N} = \{W_1, W_2, \dots, W_m\}.$$

Network Interaction

$$\mathcal{I}(\mathcal{N}) = \sum_{i,j} \mathcal{I}[W_i, W_j].$$

Networks form complex bound geometry.

Matter Structures

Matter arises through stable bound geometry.

Matter Structure

$$M(x, \Omega) = (\sigma_{AB}, L_{AB}, \mathcal{R}_{ABCD})_{\text{bound}}.$$

Matter is an emergent projection of bound geometry.

Field Structures

Field structures are projections of bound geometry after the emergence of space.

Field Projection

$$\Phi(x, \Omega) = P[C(\alpha, \Omega)].$$

Field Equation

$$\mathcal{D}\Phi = 0,$$

where $\mathcal{D}$ is an emergent differential operator.

Fields are not fundamental entities but interpretations of bound geometry.

4 Correspondence Structures

Correspondence structures connect the pre-spatial geometry of the medium with the epistemic descriptions of established physics. These correspondences are not fundamental equations but interpretative mappings that compare bound geometry—after the emergence of space—with physical concepts. Physical concepts possess no ontological status; they are appearances of bound process geometry.

Correspondence to Quantum Field Theory

Quantum Field Theory (QFT) describes fields and particles in space. In the MCM, these structures arise only through bound geometry and possess epistemic status.

Field Projection

$$\Phi(x, \Omega) = P[C(\alpha, \Omega)].$$

Correspondence

$$\Phi(x, \Omega) \leftrightarrow \text{QFT field}.$$

Particle Correspondence

$$W(\alpha, \Omega)_{\text{bound}} \leftrightarrow \text{QFT particle.}$$

QFT is an emergent description of stable vortex structures.

Correspondence to Relativity Theory

General Relativity (GR) describes spacetime curvature. In the MCM, space arises only through bound geometry and possesses no ontological fundamental status.

Emergent Metric

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Curvature Correspondence

$$\mathcal{R}_{ABCD}(\alpha, \Omega) \leftrightarrow R_{ijkl}(x, \Omega).$$

GR is a projection of internal medium curvature.

Correspondence to String Theory

String theory describes one-dimensional objects in higher-dimensional spaces. In the MCM, no fundamental spaces or strings exist; both are epistemic interpretations.

Correspondence

$$L_{AB}(\alpha, \Omega) \leftrightarrow \text{string-like modes.}$$

Interpretation Rotation generates pre-spatial patterns that may exhibit string-like behavior after projection. String theory is an interpretation of specific rotational modes.

Transition Structures

Transition structures connect unbound and bound geometry.

Temporarily Bound Geometry

$$DC(\alpha, \Omega) \approx 0 \quad (\text{temporarily}).$$

Transition

$$C(\alpha, \Omega)_{\text{open}} \rightarrow C(\alpha, \Omega)_{\text{bound}}.$$

These structures correspond to pre-material configurations.

Phase Structure of Correspondences

The correspondence structures of the PVT are phase-dependent. Each physical interpretation arises from a specific medium state. The phases determine which type of correspondence between pre-spatial geometry and epistemic physics is possible.

Phase 1: Passive Medium State In Phase 1, no internal dynamics exist. Physical concepts such as space, time, or energy possess no ontological relevance. All observable structures are projections of passive bound geometry, especially *Phase-1 islands*. These appear as stable but dynamics-free bodies resting directly on Phase 1.

Phase 2: Transition Zone Phase 2 is the dynamic transition region between passive and active medium states. Vortex spaces emerge or collapse here. Observed dynamics in astrophysical processes—such as accretion disks or jets—are manifestations of this transition phase. Black holes are not Phase 3 objects but medium states within Phase 2 where internal tension has fully collapsed.

Phase 3: Active Vortex Space In Phase 3, geometric operators act coherently. Bound vortex structures arise here and are interpreted as material appearances after the emergence of space. All correspondences to QFT and GR refer to this active phase.

Ontological Rule

Phase-3 $\rightarrow$ active correspondence, Phase-2 $\rightarrow$ transition dynamics, Phase-1 $\rightarrow$ passive projection.

Thus, every physical appearance is a phase-dependent projection of bound geometry. The PVT interprets observed dynamics not as properties of matter but as reactions of the medium to tension loss within its phase structure.

Interpretation

Correspondences are not fundamental equations but interpretative mappings. They show how bound geometry appears as physical structure after the emergence of space.

Interpretation Rule

Geometry $\rightarrow$ Physical Appearance.

All physical concepts are projections of bound geometry.

Comparison

MCM A medium, pre-spatial geometry, bound vortices.

Physics Space, time, fields, particles, forces.

Comparison Structure

Medium geometry $\leftrightarrow$ spacetime physics.

Physics is an emergent description of bound geometry.

Applications

Correspondence structures enable the interpretation of bound geometry in physical contexts.

Applications

- Interpretation of vortices as particles,
- Interpretation of rotational modes as electromagnetic fields,
- Interpretation of curvature domains as gravitational structures,
- Interpretation of networks as matter aggregates.

Correspondence Principle

The correspondence principle defines the mapping between pre-spatial geometry and physical structures.

Principle

$$\mathcal{C} : C(\alpha, \Omega) \rightarrow \text{Physics}(x, \Omega).$$

Rule

Bound Geometry $\rightarrow$ Matter and Space.

The correspondence principle is the foundation of all physical interpretations within the MCM.

5 Signatures

Within the ontological structure of the MCM, signatures are not independent physical entities but categorically defined mappings that render emergent patterns of bound and unbound medium geometry interpretable after the stabilization of spatial structure.

Signatures arise exclusively from stress, rotation, and curvature processes of the medium. They therefore possess no ontological fundamental status but represent observable projections of internal geometry after the emergence of space.

Formally, the PWT defines signatures as mappings

$$\Sigma : C(\alpha, \Omega) \rightarrow \mathcal{M}(x, \Omega),$$

where $\mathcal{M}(x, \Omega)$ denotes the class of measurable structural manifestations that exist only after the stabilization of bound geometry. Signatures are thus categorically secondary manifestations of emergent geometry.

Theoretical Predictions

Theoretical predictions of the PWT arise from analyzing the internal geometry of the medium. Each prediction is a consequence of the stress–rotation–curvature dynamics and is based on the fundamental axioms of the MCM. The following signatures are therefore not ontological entities but structural projections of bound or unbound process geometry.

Phase Dependence of Signatures

Signatures are phase-dependent manifestations of the medium. Every observable structure arises from a specific state of internal geometry. The phases determine whether a signature is active, passive, or transient.

Phase-1: Passive Signatures In Phase-1 the medium is completely quiescent. Signatures appear as stable but dynamics-free patterns of bound geometry. Typical examples are *Phase-1 islands*, whose signatures exhibit material form and mass but possess no internal vortex activity. Mathematically:

$$\mathcal{O}_L = 0, \quad DC(\alpha, \Omega) = 0.$$

Phase-2: Transitional Signatures Phase-2 produces temporary or unstable signatures. Here vortex spaces collapse or form, giving rise to short-lived resonances and high dynamical activity. Black holes are not Phase-3 objects but medium states within Phase-2, whose signatures arise from tension loss of infalling matter.

Phase-3: Active Signatures In Phase-3, stress, rotation, and curvature operators act coherently. The signatures of this phase are active and stable; they form the basis of material phenomena and spectral structures. All persistent patterns of bound geometry originate here.

Ontological Rule

Phase-3 $\rightarrow$ active signatures, Phase-2 $\rightarrow$ transitional signatures, Phase-1 $\rightarrow$ passive signatures.

Thus, signatures are not universal phenomena but phase-dependent projections of internal medium dynamics.

Signature 1: Stress-Geometric Patterns Bound vortices generate characteristic stress distributions:

$$\mathcal{O}_\sigma = \sigma_{AB}\sigma^{AB}.$$

After projection, these appear as density-like patterns in $\mathcal{M}(x, \Omega)$.

Signature 2: Rotational Modes Closed rotation produces wave-like modes:

$$\mathcal{O}_L = L_{AB}L^{AB}.$$

After the emergence of space, these correspond to electromagnetic-like structural projections.

Signature 3: Curvature Domains Coherent curvature generates large-scale domains:

$$\mathcal{O}_R = R_{ABCD}R^{ABCD}.$$

These project as gravitational or cosmological patterns.

Signature 4: Stability Indicators Stable bound geometry satisfies

$$DC(\alpha, \Omega) = 0.$$

Instability leads to transitional or decay projections.

Signature 5: Network Signatures Vortex networks generate collective patterns:

$$\Sigma_{\mathcal{N}} = \sum_{i,j} \mathcal{I}[W_i, W_j].$$

These appear as complex aggregate structures.

Signature 6: Spectral Signatures Bound vortices possess discrete eigenvalues:

$$\mathcal{O}[\Psi_n] = \lambda_n \Psi_n.$$

After projection, these correspond to spectral lines or resonances.

Signature 7: Pre-Spatial Propagation Unbound rotation produces pre-spatial propagation patterns:

$$L_{AB}(\alpha, \Omega) \rightarrow \text{Propagation}.$$

After emergence, these appear as wave phenomena.

Signature 8: Macroscopic Domains Large-scale stress organization leads to macroscopic projections:

$$\Sigma_{\text{macro}} = f(\sigma_{AB}, \mathcal{R}_{ABCD}).$$

These correspond to cosmological structures.

Signature 9: Transitional Signatures Unstable bound geometry produces temporary patterns:

$$DC(\alpha, \Omega) \approx 0 \quad (\text{temporary}).$$

These appear as short-lived resonances.

Signature 10: Matter Signatures Stable vortices generate material manifestations:

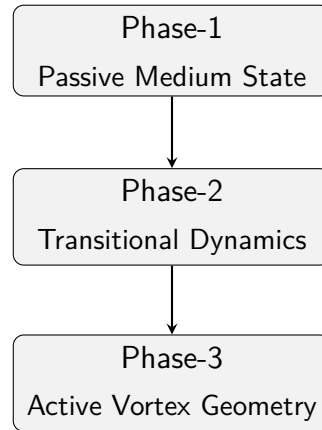
$$M(x, \Omega) = (\sigma_{AB}, L_{AB}, \mathcal{R}_{ABCD})_{\text{bound}}.$$

Matter signatures are projections of bound process geometry.

6 Structural Framework

The structural framework of the PVT defines the formal and geometric principles that determine the organization of bound and unbound geometry within the medium. It describes the categorical internal structure of the medium, the conditions for the formation of stable vortex processes, and the emergent appearances after stabilization.

The PVT distinguishes three structural levels, which are not understood as physical entities but as formal categories.



(1) Pre-spatial Structure

$$C(\alpha, \Omega) = (\rho, v_A, \sigma_{AB}, \tau)$$

denotes the internal process geometry of the medium before the emergence of space. This level has no metric, no orientation, and no physical interpretation. It represents a formal-ontological category that describes the basic structure of the medium without possessing measurable properties.

(2) Bound Structure Bound geometry arises through closed rotation, coherent curvature, and stable stress organization. The stability condition is

$$DC(\alpha, \Omega) = 0.$$

This level defines the categorical forms of bound process geometry, including vortices, clusters, and networks. It possesses ontological status within the medium but no direct measurability.

(3) Emergent Structure After stabilization, a mapping arises

$$x = X(\alpha, \Omega),$$

which makes bound geometry appear as material structure. The variable x does not denote a fundamental entity but the emergent coordinate structure arising from bound geometry. Space is therefore the external appearance of stable bound geometry and possesses no fundamental ontological status within the medium.

The structural framework connects these three levels through internal update dynamics and geometric operators that define categorical transformation rules between the levels.

Phase Architecture of the Medium

The three structural levels of the PVT correspond directly to the phases of the medium in the MCM. These phases define the ontological states in which bound and unbound geometry can occur.

Phase-1: Passive Medium State Phase 1 is the state of minimal reaction speed. The medium is completely calm and forms the ontological basis of all higher phases. It contains no active vortex coupling. *Phase-1 islands*—passive Phase 3 bodies with stable but dynamics-free structure—rest on this phase. They possess form and mass but no internal updating.

Phase-2: Transition Zone Phase 2 is the dynamic transition region between passive and active medium states. Vortex spaces emerge or collapse here. Internal updating is not neutral, leading to unstable or temporary bound states. Astrophysical phenomena such as accretion disks and jets are projections of this transition phase. Black holes are not Phase 3 objects but medium states within Phase 2 where internal tension has fully collapsed.

Phase-3: Active Vortex Space Phase 3 is the state of maximal reaction speed. Here the geometric operators act coherently and generate stable bound geometry. All vortices, clusters, and networks of the PVT exist exclusively in this phase. It forms the actual working domain of the theory.

Ontological Assignment

Phase-1 $\leftrightarrow$ passive basis, Phase-2 $\leftrightarrow$ transition dynamics, Phase-3 $\leftrightarrow$ active vortex geometry.

The phase architecture connects the structural levels of the PVT with the ontological states of the medium. Phase-1 islands mark the boundary between active vortex geometry and the passive medium state and are the only Phase 3 structures without internal dynamics.

Experimental Perspective

The experimental perspective of the PVT describes how bound and unbound geometry appear as measurable signatures after the emergence of space. This perspective is epistemic and not fundamental; it interprets emergent structures without attributing ontological status to them within the medium.

Measurable Stress Signatures

$$\mathcal{O}_\sigma = \sigma_{AB}\sigma^{AB}$$

represent the projection of stress-geometric patterns as density-like structures in the measurable domain.

Measurable Rotation Signatures

$$\mathcal{O}_L = L_{AB}L^{AB}$$

appear after projection as electromagnetic-like wave patterns.

Measurable Curvature Signatures

$$\mathcal{O}_R = R_{ABCD}R^{ABCD}$$

project as gravitational or cosmological structure patterns.

Measurable Network Signatures

$$\Sigma_{\mathcal{N}} = \sum_{i,j} \mathcal{I}[W_i, W_j]$$

appear as complex aggregate structures.

Experimental Rule

Measurement = Projection of bound geometry.

The experimental perspective is fully emergent and possesses no ontological meaning within the medium itself. It describes only the interpretation of categorical process geometry after the emergence of space.

7 Experimental Predictions

Experimental predictions of the PVT arise through the projection of bound and unbound geometry after the emergence of space. Each prediction is an interpretation of internal stress, rotation, and curvature structures of the medium. The PVT makes no physical assumptions; all experimental signatures are projections of geometric operators and possess no ontological fundamental status.

Phase Dependence of Experimental Predictions

Experimental predictions of the PVT are phase-dependent. Each observable signature arises from a specific state of the medium. The phases determine whether a prediction is active, passive, or transient.

Phase-1: Passive Predictions In Phase 1, the medium is completely calm. Predictions concern stable but dynamics-free structures such as *Phase-1 islands*. These show no internal vortex activity, only macroscopic form and mass. Observable phenomena are static and without spectral dynamics.

Phase-2: Transition Predictions Phase 2 produces dynamic, short-lived predictions. Vortex spaces emerge or collapse here, causing high energy release and light emission. Falling matter loses its tension, dissolves ontologically, and returns its energy to the medium. This energy produces the observed radiation around black holes, while the hole itself remains dark. Predictions of this phase concern tension loss, radiation, and medium reactions.

Phase-3: Active Predictions In Phase 3, geometric operators act coherently. All stable vortex structures, spectral lines, and resonance patterns arise here. Predictions of this phase are active and persistently measurable; they form the basis of all experimentally testable signatures of the PVT.

Ontological Rule

Phase-3 $\rightarrow$ active predictions, Phase-2 $\rightarrow$ transition predictions, Phase-1 $\rightarrow$ passive predictions.

Thus, experimental predictions are not universal phenomena but phase-dependent projections of internal medium dynamics.

Spectral Predictions

Bound vortices possess discrete eigenvalues of the internal geometry:

$$\mathcal{O}[\Psi_n] = \lambda_n \Psi_n.$$

Prediction Discrete eigenvalues project after emergence as spectral lines or resonances in the measurable domain.

Spectral Signature

$$\lambda_n \in \mathbb{R}_{\text{discrete}}.$$

Unstable structures possess a continuous spectrum that appears as blurred or broadened signatures.

Resonance Predictions

Rotational modes of bound vortices generate characteristic resonance patterns.

Rotation Observable

$$\mathcal{O}_L = L_{AB} L^{AB}.$$

Prediction Closed rotation leads to discrete resonance frequencies that appear after emergence as electromagnetic-like structural projections.

Resonance Condition

$$\oint L_{AB} d\alpha^A = n \Phi_0.$$

Propagation Predictions

Unbound rotation generates pre-spatial propagation patterns.

Prediction

$$L_{AB}(\alpha, \Omega) \rightarrow \text{Propagation.}$$

After emergence, these patterns appear as wave-like projections.

Propagation Structure

$$\mathcal{P} = f(L_{AB}, \tau).$$

Field Predictions

Bound geometry projects after emergence as field-like structure.

Field Projection

$$\Phi(x, \Omega) = P[C(\alpha, \Omega)].$$

Prediction Fields are not fundamental entities but interpretations of bound stress, rotation, and curvature geometry.

Field Signature

$$\mathcal{D}\Phi = 0.$$

Topological Predictions

Topological structures of the medium generate characteristic projections.

Topological Charge

$$\mathcal{Q} = \frac{1}{\Phi_0} \oint L_{AB} d\alpha^A.$$

Prediction Topological charges appear after emergence as quantized parameters (e.g., charge, spin) without being fundamental entities themselves.

Quantization

$$\mathcal{Q} \in \mathbb{Z}.$$

Macroscopic Predictions

Large-scale stress and curvature organization generates macroscopic structural projections.

Macroscopic Structure

$$\Sigma_{\text{macro}} = f(\sigma_{AB}, \mathcal{R}_{ABCD}).$$

Prediction Macroscopic signatures appear as large-scale matter or field distributions.

Cosmological Predictions

Coherent curvature domains generate cosmological projections.

Curvature Observable

$$\mathcal{O}_R = R_{ABCD}R^{ABCD}.$$

Prediction Large-scale curvature organization appears as cosmological structures (e.g., background signatures, domain formation).

Domain Structure

$$\mathcal{D}_{\text{cosmo}} = f(\mathcal{R}_{ABCD}).$$

Summary

Experimental predictions of the PVT are projections of internal geometry. They include:

- discrete spectra,
- rotational resonances,
- propagation patterns,
- field-like structures,
- topological quantization,
- macroscopic patterns,
- cosmological domains.

All predictions are geometric appearances of bound or unbound process geometry.

Indirect Measurability

Many signatures of the PVT are indirectly measurable because they appear only after the emergence of space.

Indirect Measurement Rule

Measurement = Projection of bound geometry.

Indirect Signatures

- spectral lines,
- resonance frequencies,
- field-like distributions,
- macroscopic patterns,
- cosmological structures.

Macroscopic Signatures

Macroscopic signatures arise through large-scale stress and curvature organization.

Signature

$$\Sigma_{\text{macro}} = f(\sigma_{AB}, \mathcal{R}_{ABCD}).$$

Prediction These signatures appear as large-scale matter or field distributions (e.g., clusters, domains, networks).

Network Signature

$$\Sigma_{\mathcal{N}} = \sum_{i,j} \mathcal{I}[W_i, W_j].$$

8 Known Data Relevant to the PVT

Known data from established physics are not used in the PVT as fundamental quantities but as projections of bound geometry after the emergence of space. The PVT interprets experimental data exclusively as signatures of internal stress, rotation, and curvature structures of the medium. They therefore possess epistemic, not ontological, status.

This chapter assigns known experimental data to the corresponding geometric structures of the PVT.

Phase-Dependent Interpretation of Known Data

Known experimental data of physics are phase-dependent appearances of the medium. Their observed forms and signatures arise from specific states of the internal geometry and can be assigned to the three phases of the medium.

Phase-1: Passive Data In Phase 1, the medium is completely calm. Observable structures are stable but dynamics-free. Typical examples are *Phase-1 islands*, whose data show form, mass, and static field distributions but no internal vortex activity. These data appear as stationary macroscopic patterns.

Phase-2: Transition Data Phase 2 produces dynamic, short-lived data. Vortex spaces emerge or collapse here, causing high energy release and light emission. Astrophysical observations such as accretion disks, jets, and the apparent form of black holes are projections of this transition phase. Falling matter loses its tension, dissolves ontologically, and returns its energy to the medium. The released energy produces the observed radiation, while the black hole itself remains dark.

Phase-3: Active Data In Phase 3, geometric operators act coherently. All stable vortex structures, spectral lines, and resonance patterns arise here. Data of this phase are active and persistently measurable; they form the basis of all experimentally verifiable signatures of the PVT.

Ontological Rule

Phase-3 $\rightarrow$ active data, Phase-2 $\rightarrow$ transition data, Phase-1 $\rightarrow$ passive data.

Thus, known experimental data are not universal measurement quantities but phase-dependent projections of internal medium dynamics.

Spectral Data

Spectral data in physics correspond to discrete eigenvalues of bound process geometry.

Geometric Basis

$$\mathcal{O}[\Psi_n] = \lambda_n \Psi_n.$$

Interpretation Discrete spectra (e.g., atomic lines) are projections of discrete eigenmodes of bound vortices.

Signature

$$\lambda_n \in \mathbb{R}_{\text{discrete}}.$$

Spectral data indicate bound stress, rotation, and curvature geometry.

Interference and Propagation Data

Interference patterns and propagation phenomena arise from unbound rotational modes of the medium.

Geometric Basis

$$L_{AB}(\alpha, \Omega) \rightarrow \text{Propagation}.$$

Interpretation Interference patterns are projections of superposed rotational modes.

Propagation Structure

$$\mathcal{P} = f(L_{AB}, \tau).$$

These data reveal the pre-spatial propagation dynamics of the medium.

Field and Coherence Data

Field structures and coherence phenomena are projections of bound stress, rotation, and curvature geometry.

Field Projection

$$\Phi(x, \Omega) = P[C(\alpha, \Omega)].$$

Coherence Coherent rotational modes generate coherent field-like structures.

Signature

$$\mathcal{O}_L = L_{AB}L^{AB}.$$

Field and coherence data indicate bound rotational geometry.

Macroscopic and Network Data

Macroscopic matter structures and network phenomena arise from vortex clusters and vortex networks.

Network Definition

$$\mathcal{N} = \{W_1, W_2, \dots, W_m\}.$$

Network Signature

$$\Sigma_{\mathcal{N}} = \sum_{i,j} \mathcal{I}[W_i, W_j].$$

Macroscopic Structure

$$\Sigma_{\text{macro}} = f(\sigma_{AB}, \mathcal{R}_{ABCD}).$$

Macroscopic data show the collective organization of bound geometry.

Cosmological Data

Cosmological data correspond to large-scale curvature domains of the medium.

Curvature Observable

$$\mathcal{O}_R = R_{ABCD} R^{ABCD}.$$

Domain Structure

$$\mathcal{D}_{\text{cosmo}} = f(\mathcal{R}_{ABCD}).$$

Interpretation Cosmological background signatures are projections of coherent curvature organization. They reveal the large-scale structure of the medium.

Summary

Known experimental data of physics are projections of internal medium geometry:

- spectral data $\rightarrow$ discrete eigenmodes of bound vortices,
- interference data $\rightarrow$ superposed rotational modes,
- field data $\rightarrow$ projections of bound geometry,
- network data $\rightarrow$ vortex clusters and networks,
- cosmological data $\rightarrow$ large-scale curvature domains.

All data are emergent appearances of a single medium and possess no ontological fundamental status.

9 Comparison with Established Physics

The comparison between the Planck–Vortex Theory (PVT) and established physics takes place exclusively at the level of emergent appearances. The PVT introduces no physical fundamental entities. Space, time, fields, particles, and forces arise only through bound geometry of the medium and therefore possess epistemic, not ontological, status.

Established physics describes observable phenomena in space. The PVT describes the pre-spatial process geometry of the medium from which these phenomena emerge. A direct comparison is therefore possible only as a correspondence between *categorical geometric structures* and *epistemic physical appearances*.

Phase-Dependent Comparison Level

The comparison between the PVT and established physics is phase-dependent. The three phases of the medium determine which type of physical appearance is observable and how it is interpreted in established physics.

Phase-1: Passive Appearances In Phase 1, the medium is completely calm. Physical concepts such as mass, shape, and gravitation appear as static projections of bound geometry. *Phase-1 islands* correspond to classical bodies of mechanics whose dynamics are purely external. Classical physics describes these appearances as motion in space, while the PVT interprets them as geometrically bound but internally neutral structures.

Phase-2: Transition Appearances Phase 2 produces dynamic, short-lived appearances. Vortex spaces emerge or collapse here, causing high energy release and light emission. Astrophysical phenomena such as accretion disks, jets, and the observed form of black holes are projections of this phase. Relativity theory describes them as extreme spacetime curvature, while the PVT interprets them as tension loss and medium reaction.

Phase-3: Active Appearances In Phase 3, geometric operators act coherently. All stable vortex structures, particles, and fields of quantum field theory arise here. The PVT describes these appearances as bound process geometry that appears as material structure after the emergence of space. Quantum mechanics is therefore an epistemic description of active vortex spaces.

Ontological Rule

Phase-3 ↔ quantum physics, Phase-2 ↔ relativistic dynamics, Phase-1 ↔ classical mechanics.

Thus, the phase-dependent comparison shows that established physics is not contradictory to the PVT but describes different projections of the same medium geometry.

Comparison Structure

The comparison structure is based on the mapping

$$\mathcal{C} : C(\alpha, \Omega) \rightarrow \mathcal{M}(x, \Omega),$$

which transforms bound geometry of the medium into measurable appearances. $\mathcal{M}(x, \Omega)$ denotes the class of epistemic structures that exist only after the emergence of space.

The PVT describes:

- stress geometry,
- rotational geometry,
- internal curvature,
- update dynamics,
- bound vortex structures.

Established physics describes:

- fields,
- particles,
- forces,
- spacetime,
- interactions.

These two descriptions are not equivalent but stand in an emergence relation.

Comparison with Quantum Field Theory

QFT describes fields and particles in space. In the PVT, these structures arise only through bound geometry.

Field Correspondence

$$\Phi(x, \Omega) = P[C(\alpha, \Omega)].$$

Particle Correspondence

$$W(\alpha, \Omega)_{\text{bound}} \leftrightarrow \text{QFT particle.}$$

Comparison

- QFT fields are projections of bound stress, rotation, and curvature geometry.
- QFT particles are epistemic interpretations of stable vortex structures.
- QFT interactions are superpositions of bound geometry.

Comparison with Relativity Theory

General Relativity describes spacetime curvature. In the PVT, space arises only through bound geometry.

Emergent Metric

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Curvature Correspondence

$$\mathcal{R}_{ABCD}(\alpha, \Omega) \leftrightarrow R_{ijkl}(x, \Omega).$$

Comparison

- GR spacetime is a projection of bound geometry.
- Gravitation is emergent curvature organization.
- Geodesics are projections of internal update dynamics.

Comparison with String Theory

String theory describes one-dimensional objects in higher-dimensional spaces. The PVT contains no fundamental spaces or strings.

Rotation Correspondence

$$L_{AB}(\alpha, \Omega) \leftrightarrow \text{string-like modes.}$$

Comparison

- Strings are epistemic interpretations of specific rotational modes.
- The PVT requires no additional dimensions.
- String oscillations correspond to rotational modes of bound vortices.

Comparison with Classical Physics

Classical physics describes forces, fields, and motion in space. The PVT describes internal geometric relations within the medium.

Comparison

- Forces are emergent interactions of bound geometry.
- Motion is a projection of internal updating.
- Fields are projections of bound stress, rotation, and curvature structures.

Summary Comparison

The PVT is not an alternative physical theory but a pre-spatial geometric theory that interprets physics as emergent structure.

Ontological Relation

Medium geometry $\rightarrow$ physical appearance.

Key Differences

- PVT: one medium, pre-spatial geometry, vortices.
- Physics: space, time, fields, particles, forces.

Key Correspondences

- vortices $\leftrightarrow$ particles,
- rotational modes $\leftrightarrow$ electromagnetic fields,
- curvature domains $\leftrightarrow$ gravitational structures,

- networks $\leftrightarrow$ matter aggregates.

Physics is the external appearance of bound geometry.

10 QM–ART Correspondence

The QM–ART correspondence of the PVT describes the shared geometric foundation of Quantum Mechanics (QM) and General Relativity (GR). Both theories arise as different projections of the same pre-spatial process geometry of the medium and therefore possess epistemic, not ontological, status.

The PVT shows that QM and GR are not fundamental theories but emergent descriptions of bound geometry. The correspondence arises through the categorical mapping

$$\mathcal{C} : C(\alpha, \Omega) \rightarrow \{\text{QM}, \text{GR}\},$$

which transforms internal process geometry into different epistemic appearances. QM and GR are two limiting cases of the same update order.

Discrete Configuration Space

The configuration space of the PVT is the medium-index space

$$\alpha \in \mathcal{A},$$

which possesses no spatial structure. All configurations are discrete states of the medium and have no physical status.

Configuration Set

$$V = \{v_i \mid i \in I\},$$

where I is a discrete index set.

Hilbert Space of the PVT

$$H_{\text{PVT}} = \ell^2(V),$$

with basis vectors

$$|v\rangle, \quad v \in V.$$

This Hilbert space is a formal category and has no physical interpretation. It serves solely to describe the structural aspects of pre-spatial process geometry.

Absence of Fundamental Time

The PVT contains no fundamental time variable. The internal evolution of the medium is described by the update index Ω .

Update Dynamics

$$DC(\alpha, \Omega)$$

defines the internal process evolution.

QM Time

$$t_{\text{QM}} = \lim_{\Delta\Omega \rightarrow 0} \Omega$$

is a continuum limit of the update order and has no ontological status.

GR Time

$$t_{\text{GR}} = \lambda$$

is a limiting case of geodesic propagation order. Time is emergent in both cases.

QM Correspondence

Quantum Mechanics describes the spectral structure of bound geometry. In the PVT, QM arises as the limiting case of the spectral operator structure of the medium.

Spectral Operator

$$\mathcal{O}[\Psi_n] = \lambda_n \Psi_n.$$

Correspondence

$$\Psi(\alpha, \Omega) \leftrightarrow \psi(x, t).$$

Schrödinger Equation as a Limit

$$i \frac{\partial}{\partial t} \psi(x, t) = H_{\text{QM}} \psi(x, t)$$

corresponds to the continuum limit of internal updating

$$DC(\alpha, \Omega).$$

QM is the spectral projection of bound geometry.

GR Correspondence

General Relativity describes spacetime curvature. In the PVT, space arises only through bound geometry.

Emergent Metric

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Geodesics

$$\frac{d^2 x^i}{d\lambda^2} + \Gamma_{jk}^i \frac{dx^j}{d\lambda} \frac{dx^k}{d\lambda} = 0.$$

Curvature Correspondence

$$\mathcal{R}_{ABCD}(\alpha, \Omega) \leftrightarrow R_{ijkl}(x, \Omega).$$

GR is the geodesic projection of bound curvature domains.

Unified Structure

QM and GR arise from the same pre-spatial process geometry of the medium.

Unified Operator

$$H_{\text{PVT}} = H_{\text{geom}}[\gamma_{AB}] + H_{\text{int}}[\sigma_{AB}, L_{AB}].$$

Shared Foundation

- QM: spectral structure of bound geometry,
- GR: geodesic structure of bound geometry.

Both theories are limiting cases of the same update order.

Cross-Phase Correspondence

The QM–GR correspondence spans multiple phases. Both theories arise from different projections of the same medium geometry but in different phases of the medium state.

Phase-3: Quantum Projection In Phase 3, internal geometry is active. The spectral structure of bound vortex spaces generates the appearances of QM. Operators such as $\mathcal{S}$, $\mathcal{L}$, and $\mathcal{R}$ act coherently and form discrete eigenmodes. QM is therefore the epistemic projection of active Phase 3.

Phase-2: Relativistic Projection In Phase 2, the medium is in transition between active and passive geometry. Large-scale curvature domains and tension gradients arise here, appearing after emergence as spacetime curvature. GR is the epistemic projection of this transition phase. Black holes are not Phase 3 objects but medium states within Phase 2 where tension has fully collapsed.

Phase-1: Geometric Basis Phase 1 forms the resting foundation of both projections. It contains no active dynamics but supports the passive structures of higher phases. Phase-1 islands are geometric boundary cases between active vortex geometry and the passive medium state.

Ontological Rule

Phase-3 $\leftrightarrow$ QM, Phase-2 $\leftrightarrow$ GR, Phase-1 $\leftrightarrow$ medium basis.

Thus, the cross-phase correspondence shows that QM and GR are not separate theories but two appearances of the same pre-spatial medium geometry in different phases of the update order.

Summary

The QM–GR correspondence shows:

- QM and GR are not fundamental theories,
- both arise from bound stress, rotation, and curvature geometry,
- both use different projections of the same structure,
- time is emergent in both cases,
- the configuration space is pre-spatial and discrete,
- the PVT unifies QM and GR geometrically.

QM and GR are two epistemic appearances of a single medium.

11 Formal Consistency

The formal consistency of the Planck–Vortex Theory (PVT) ensures that all equations, operators, topological structures, and dynamic processes are fully compatible with the ontology of the Monistic Continuum Model (MCM). Formal consistency means that all structures arise exclusively from internal geometric relations of the medium and introduce no additional ontological requirements. Formal consistency comprises six categorical levels:

- ontological consistency,
- operatorial consistency,
- topological consistency,
- dynamic consistency,
- spectral consistency,
- emergent consistency.

Ontological Consistency

Ontological consistency requires that all structures of the PVT arise exclusively from the medium. No physical fundamental entities may be introduced; all appearances are projections of bound geometry.

Ontological Condition

Ontology: Only Medium.

State Mapping

$$C(\alpha, \Omega) = (\rho, v_A, \sigma_{AB}, \tau).$$

Consistency Criteria

- no spatial coordinates,
- no time variable,
- no fields,
- no particles,

- no forces.

All structures must be purely geometric relations of the medium.

Operatorial Consistency

Operatorial consistency requires that all operators of the PVT are purely geometric and act exclusively on medium indices. They possess no physical status and are categorically closed.

Geometric Operators

$$\mathcal{S}[\sigma_{AB}], \quad \mathcal{L}[v_A], \quad \mathcal{R}[\gamma_{AB}], \quad D[C].$$

Consistency Rule

$$\mathcal{O} : \mathcal{A} \rightarrow \mathcal{A}.$$

Prohibition

- no physical operators,
- no differential operators over spacetime,
- no field operators.

All operators must be geometrically closed.

Topological Consistency

Topological consistency requires that rotational structures form closed loops and that the binding of a vortex possesses quantized topological charge.

Topological Binding Condition

$$\oint L_{AB} d\alpha^A = n \Phi_0, \quad n \in \mathbb{Z}.$$

Topological Charge

$$\mathcal{Q} = \frac{1}{\Phi_0} \oint L_{AB} d\alpha^A.$$

Topological consistency is necessary for stable vortices.

Dynamic Consistency

Dynamic consistency requires that the internal updating of the medium satisfies the stability conditions of bound geometry.

Update Operator

$$DC(\alpha, \Omega).$$

Stability Condition

$$DC(\alpha, \Omega) = 0.$$

Instability

$$DC(\alpha, \Omega) \neq 0.$$

Stress Dynamics

$$D\sigma_{AB} = F(Vv, \sigma)_{AB}.$$

Dynamic consistency is the foundation of bound geometry.

Spectral Consistency

Spectral consistency requires that bound vortices possess discrete eigenvalues and unstable structures possess continuous spectra.

Spectral Operator

$$\mathcal{O}[\Psi_n] = \lambda_n \Psi_n.$$

Stable

$$\lambda_n \in \mathbb{R}_{\text{discrete}}.$$

Unstable

$$\lambda \in \mathbb{R}_{\text{continuous}}.$$

Spectral consistency is necessary for material stability.

Emergent Consistency

Emergent consistency requires that the projection of bound geometry is consistent with physical appearances. The projection possesses epistemic status and is not an ontological structure.

Emergent Mapping

$$x = X(\alpha, \Omega).$$

Emergent Metric

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Emergent Rule

Bound Geometry $\rightarrow$ Matter and Space.

Emergent consistency connects the PVT with observable physics.

Phase-Dependent Consistency

Phase-dependent consistency requires that all equations and operators of the PVT remain valid in each phase of the medium without generating ontological contradictions. The phases define the validity domains of the operators and the stability conditions of bound geometry.

Phase-1: Passive Consistency Domain In Phase 1, the medium is completely calm. All operators are stationary, and the equations possess trivial solutions:

$$DC(\alpha, \Omega) = 0, \quad \mathcal{L}[v_A] = 0.$$

Phase-1 islands are geometrically bound but dynamically neutral. They satisfy all consistency conditions without internal updating.

Phase-2: Transition Consistency In Phase 2, internal updating is not neutral. The equations are unstable, and the operators generate open or collapsing geometries. Consistency in this phase arises from the preservation of the topological binding condition despite tension loss:

$$\oint L_{AB} d\alpha^A = n \Phi_0, \quad n \in \mathbb{Z}.$$

Black holes are medium states of this phase in which tension has fully collapsed while topological order remains intact.

Phase-3: Active Consistency Domain In Phase 3, all operators act coherently. The equations are closed, and the stability conditions are satisfied:

$$DC(\alpha, \Omega) = 0, \quad \lambda_n \in \mathbb{R}_{\text{discrete}}.$$

This phase forms the operational domain of the PVT and ensures full formal consistency of the theory.

Ontological Rule

Phase-3 $\rightarrow$ active consistency, Phase-2 $\rightarrow$ transition consistency, Phase-1 $\rightarrow$ passive consistency.

Thus, the formal consistency of the PVT is not only mathematical but also ontological: all equations remain closed and free of contradictions within the phase structure of the medium.

Summary

Formal consistency of the PVT requires:

- ontological purity,
- geometric operators,
- closed rotation,
- stable updating,
- discrete spectra,
- consistent projection.

The PVT is fully consistent when all structures arise exclusively from the medium and all equations are geometrically closed.

12 Summary

The Planck–Vortex Theory (PVT) is a fully geometric theory that interprets all observable phenomena as emergent appearances of a single medium. The PVT is based on the ontological axioms of the Monistic Continuum Model (MCM) and uses only internal stress, rotation, and curvature geometry. All observable structures possess epistemic status and arise only after projection of bound geometry.

The theory shows:

- the medium is the only fundamental entity,
- all structures are pre-spatial process geometries,

- bound geometry forms the basis of epistemic matter and spatial appearances,
- vortices are the only stable bound configurations,
- physical concepts (space, time, fields, particles, forces) are projections of bound geometry,
- QM and GR are limiting cases of the same geometric structure,
- experimental signatures are appearances of internal operators.

The PVT thus provides a consistent geometric framework that describes physics as an emergent appearance of a single medium.

Future Directions

The future development of the PVT encompasses several categorical research fields:

(1) Pre-spatial Spectral Structures Deepening the analysis of discrete and continuous spectra of bound geometry, especially higher-order vortices and networks.

(2) Global Curvature Domains Investigation of large-scale curvature organization and its projection into cosmological appearances.

(3) Dynamic Transition Structures Analysis of unstable bound geometry and pre-material transitions:

$$DC(\alpha, \Omega) \approx 0 \quad (\text{temporarily}).$$

(4) Emergence Mechanisms Deepening the mapping

$$x = X(\alpha, \Omega),$$

which generates epistemic spatial and material structure.

(5) Correspondence Expansion Further development of the QM–GR correspondence and investigation of additional physical theories as projections of bound geometry.

(6) Network Physics Analysis of complex vortex networks and their emergent macroscopic signatures.

These directions form the basis for a comprehensive geometric description of epistemic physical appearances.

Conclusion

The Planck–Vortex Theory shows that all observable phenomena emerge from the internal geometry of a single medium. The PVT does not replace physics but explains it as a projection of bound geometry.

Central Conclusion

Matter = bound process geometry, Space = emergent projection.

Fundamental Relation

$$C(\alpha, \Omega) \rightarrow M(x, \Omega).$$

The PVT thus provides a consistent, ontologically minimal, and formally closed theory that describes physics as the external appearance of a single medium. It unifies spectral, topological, dynamic, and geometric structures into a single framework.

The PVT is therefore a fully monistic model forming the foundation for a geometric unification of epistemic physical appearances.

Phase Integration of the Theory

The full integration of the phase structure into the PVT shows that all ontological, mathematical, and experimental levels of the theory are coherently connected. The three phases of the medium form the fundamental basis of all appearances:

Phase-1: Passive Medium State The medium is completely calm and forms the ontological basis. *Phase-1 islands* are passive Phase 3 bodies with stable but dynamics-free structure. They possess form and mass but no internal updating.

Phase-2: Transition Zone In this phase, vortex spaces emerge or collapse. It is the dynamic bridge between passive and active medium states. Astrophysical appearances such as accretion disks and black holes are projections of this transition phase. The internal tension of the medium collapses, releasing energy.

Phase-3: Active Vortex Space Here the geometric operators act coherently. All stable vortices, clusters, and networks exist exclusively in this phase. It forms the operational domain of the PVT and generates the observable signatures of physics.

Ontological Rule

Phase-3 $\rightarrow$ active vortex geometry, Phase-2 $\rightarrow$ transition dynamics, Phase-1 $\rightarrow$ passive basis.

Thus, the PVT is fully phase-integrated: all appearances of physics are projections of bound geometry within the phase structure of the medium. The theory remains ontologically minimal, formally closed, and experimentally interpretable.

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